Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you need more room, use the backs of the pages and indicate that you have done so.
- If you have a question, raise your hand and I will come to you.
- You have 50 minutes to complete the exam.
- Good luck!

Question:	1	2	3	4	5	Total
Points:	10	10	10	10	10	50
Score:						

1. (10 points) For each of the following statements, write the negation of the statement without using the logical operator "$\neg$" or the word "not." You can use either symbolic logic or an English sentence. Determine whether the original statement or its negation is true, and give a proof of the true statement.

   (a) $(\forall n \in \mathbb{Z})(\exists m \in \mathbb{Z})(n + m = -n)$.

   (b) There exists a real number x such that, for any real number y, if $x + y > 0$ then $y > 0$.

2. (10 points) Let P, Q and R be statements. Show that the following logical equivalences hold.

   (a) $\big((P \to Q) \wedge (R \to Q)\big) \equiv \big((P \vee R) \to Q\big)$.

   (b) $\neg(P \leftrightarrow Q) \equiv (P \wedge \neg Q) \vee (Q \wedge \neg P)$.

3. (10 points) Decide whether the following statements are true or false. If the statement is true, prove it. If it is false, give a counterexample.

   (a) Let A, B and C be sets. Then $(A \backslash B) \cap (A \backslash C) = A \backslash (B \cap C)$.

   (b) The set $A = \{7m \mid m \in \mathbb{Z}\}$ is closed under addition.

4. (10 points) Show that $3^n > 2^n + 1$ for every integer $n \geqslant 2$.

5. (10 points) Recall that the Fibonacci numbers are defined by $F_0 = 1$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geqslant 2$.

   Prove that $\sum_{i=0}^{n} F_i = F_{n+2} - 1$ for every integer $n \geqslant 0$.

Axioms

Field axioms

1. **Closure:** If $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
2. **Commutativity:** If $a, b \in \mathbb{R}$, then $a + b = b + a$ and $ab = ba$.
3. **Associativity:** If $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
4. **Distributivity:** If $a, b, c \in \mathbb{R}$, then $(a+b)c = ac+bc$ and $a(b + c) = ab + ac$.
5. **Existence of Identities:** There are two different real numbers 0 and 1 such that for all real numbers a, $a + 0 = a$ and $a \cdot 1 = a$.
6. **Additive Inverses:** For all $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
7. **Multiplicative Inverses:** For all $a \in \mathbb{R}$ such that $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

- $\mathbb{N} \cup \{0\}$ satisfies Axioms 1–5 (quantified over $\mathbb{N}$)
- $\mathbb{Z}$ satisfies Axioms 1–6 (quantified over $\mathbb{Z}$)
- $\mathbb{Q}$ satisfies Axioms 1–7 (quantified over $\mathbb{Q}$)

Equality Axioms. These hold for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

8. **Reflexivity:** For every real number a, $a = a$.
9. **Symmetry:** For all real numbers a and b, if $a = b$, then $b = a$.
10. **Transitivity:** For all real numbers a, b, and c, if $a = b$ and $b = c$ then $a = c$.
11. **Substitution:** If $a = b$, then b may be substituted for a in any statement.

Order Axioms. We assume $\mathbb{R}$ contains a subset $\mathbb{R}^+$ of "positive real numbers" such that

12. **Positive Closure:** If $a, b \in \mathbb{R}^+$, then $a + b \in \mathbb{R}^+$ and $ab \in \mathbb{R}^+$.
13. **Trichotomy:** If $a \in \mathbb{R}$, then exactly one of the following is true:

    • $a \in \mathbb{R}^+$ • $-a \in \mathbb{R}^+$ • $a = 0$.

14. **Well-Ordering Property:** If S is a nonempty set of positive integers, then S has a minimum element.

Facts

Logic. Let P, Q and R be statements.

- $P \to Q \equiv \neg P \vee Q$
- $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$

De Morgan's Laws

- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Distributive Laws

- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Sets. Let A and B be subsets of universe U.

- $A \subseteq B$ if $x \in A$ implies $x \in B$.
- $A = B$ if $A \subseteq B$ and $B \subseteq A$.
- $A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$
- $A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$
- $A \backslash B = \{x \in U : x \in A \text{ and } x \notin B\}$
- $A^c = \{x \in U : x \notin A\}$

Inequalities. Let $a, b \in \mathbb{R}$.

- We say $a < b$ if $b - a \in \mathbb{R}^+$.
- We say $a \leqslant b$ if $a < b$ or $a = b$.
- We say $a > b$ if $a - b \in \mathbb{R}^+$.
- We say $a \geqslant b$ if $a > b$ or $a = b$.

Real numbers. Let $a, b, c \in \mathbb{R}$.

R1 If $a = b$ then $a + c = b + c$ and $ac = bc$.
R2 If $a + c = b + c$ then $a = b$.
R3 The additive identity in $\mathbb{R}$ is unique.
R4 $0 \cdot a = 0$.
R5 $(-1)a = -a$.
R6 $(-1)(-1) = 1$
R7 $(-a)(-b) = ab$
R8 $-(ab) = (-a)b = a(-b)$.
R9 1 is positive.
R10 $a \in \mathbb{R}^+$ if and only if $a > 0$.
R11 if $a > b$ and $c > 0$ then $ac > bc$.
R12 if $a > b$ then $a + c > b + c$.